

AIKR - S2026 Project

1. Project Objective

The objective of this project is to design and implement an **AI Agent from scratch** that can **perceive inputs, reason, and perform actions to achieve a defined goal**.

Students are encouraged to apply concepts learned in **Artificial Intelligence such as search, reasoning, planning, machine learning, or decision making**.

The agent may be implemented in **any programming language** (Python preferred).

Students may optionally design a **User Interface (UI)** to interact with the agent.

2. Project Requirements

Mandatory Requirements

Each project must include the following components:

1. Problem Definition

- Clearly define the problem your AI Agent solves.
- Explain why an AI agent approach is suitable.

2. Agent Design

- Define:
 - Environment
 - Agent type (simple reflex / goal-based / utility-based / learning agent)
 - Inputs (Percepts)
 - Outputs (Actions)

3. AI Logic

The agent must include **at least one AI technique**, such as:

- Search algorithms (BFS, DFS, A*, etc.)
- Planning
- Knowledge representation
- Rule-based reasoning
- Machine learning

4. Implementation

- Code implementation of the agent.
- Demonstration of the agent performing tasks.

5. Testing

- Demonstrate at least **3 test scenarios**.
 - Show performance or decision-making behaviour.
-

3. Use of AI Tools (Important)

Students are allowed to use **AI tools** such as:

- ChatGPT
- GitHub Copilot
- Claude
- Gemini
- Code generation tools
- UI builders

However:

Rules

1. **All AI tool usage must be clearly acknowledged.**
2. Students must **understand and explain all generated code**.
3. Copying entire projects from AI or GitHub without understanding is **not allowed**.
4. Students who build **major components from scratch** will receive **higher marks**.

AI Usage Declaration (Mandatory)

Students must include a section in the report:

AI Assistance Declaration

Example:

- AI Tool Used: ChatGPT
- Purpose: Helped debug Python code and suggest UI layout
- AI Generated Components:
 - Login page layout
 - Debugging suggestions

Students must clearly state **what was AI-generated vs self-written**.

Failure to declare AI usage may result in **zero marks**.

4. Optional Enhancements (Bonus)

Students can improve their projects by adding:

- Graphical **User Interface (UI)**
 - Visualization of agent decisions
 - Real-world dataset integration
 - Performance optimization
 - Multi-agent interaction
 - Deployment (web app / API)
-

5. Deliverables

Students must submit the following:

1. Project Report (PDF)

Maximum **8–10 pages**

Must include:

1. Problem Statement
 2. Agent Architecture
 3. AI Techniques Used
 4. System Design
 5. Implementation Details
 6. Results & Testing
 7. Limitations
 8. AI Tool Usage Declaration
 9. References
-

2. Source Code

Submit a **compressed folder (.zip)** containing:

ProjectName/
|
|— source_code/
|— dataset (if any)
|— README.md
|— requirements.txt

The **README** file must include:

- Project description
 - Installation steps
 - How to run the project
 - Example input/output
-

3. Demo Video (3–5 minutes)

The video should demonstrate:

- Problem explanation
 - System working
 - Agent making decisions
 - UI interaction (if available)
-

6. Evaluation Rubric (100 Marks)

Component	Marks
Problem Definition & Idea	10
Agent Design & Architecture	10
AI Logic / Algorithms	20
Implementation Quality	20
Testing & Demonstration	10
Report Quality	10

Originality & From-Scratch Work	20
---------------------------------	----

Bonus

Component	Bonus
UI Development	+5
Innovative Features	+5

7. Academic Integrity

Students must adhere to **academic honesty**.

The following are **not allowed**:

- Submitting copied GitHub projects
- Submitting AI-generated full solutions without understanding
- Plagiarism from other teams

Violation may lead to Zero marks

8. Submission Instructions

Submit all materials on **AIKR Google Classroom** by 04th May 2026.

Submission Format: RollNumber_Name_AIKRProject.zip

9. Demonstration / Viva

Students must be able to explain:

- Agent architecture
- Algorithms used
- Code structure
- AI tool usage
- Future improvements